INCH-POUND
MIL-PRF-1/36G
16 December 2002
SUPERSEDING
MIL-PRF-1/36F
22 July 1997

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBES, CATHODE RAY

TYPES 3JP1, 3JP2, 3JP7, 3JP11, AND 3JP12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus.

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

Parameter:	Type	EF	EC1	ed	Eb1	Eb2	Eb3	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:		V	V dc	v	V dc	V dc	V dc	Meg	Meg	V dc	Ratio	ft
Maximum:	P1	6.9	0	550	1,100	2,200	4,400	1.5	1.5	-125	2.3	50,000
	P2	6.9	0	550	1,100	2,200	4,400	1.5	1.5	-125	2.3	50,000
	P11	6.9	0	550	1,100	2,200	4,400	1.5	1.5	-125	2.3	50,000
	P7	6.9	0	550	1,100	2,200	4,400	1.5	1.0	-125	2.3	50,000
	P12	6.9	0	550	1,100	2,200	4,400	1.5	1.0	-125	2.3	50,000
Minimum:	All:	5.7	-200	- - -	- - -	1,500	3,000	- - -	- - -	- - -	- - -	- - -
Test Conditions:												
	P1	6.3	Adj	- - -	Focus	1,500	3,000	- - -	- - -	- - -	- - -	- - -
	P2	6.3	Adj	- - -	Focus	1,500	3,000	- - -	- - -	- - -	- - -	- - -
	P11	6.3	Adj	- - -	Focus	1,500	3,000	- - -	- - -	- - -	- - -	- - -
	P12	6.3	Adj	- - -	Focus	1,500	3,000	- - -	- - -	- - -	- - -	- - -
	P7	6.3	Adj	- - -	Focus	2,000	4,000	- - -	- - -	- - -	- - -	- - -

GENERAL:

Qualification - Required

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Table 1. Testing and Inspection.

Method	Requirement or Test	Type	Conditions	Symbol	Limits Min	Limits Max	Unit
	Qualification inspection						
1002	Barometric pressure, reduced	All	87.0 mm Hg		- - -	- - -	- - -
5101	Neck and bulb alignment (electrostatic types)	All		dia	- - -	2.25	in.
5246	Focusing voltage, zero bias	P1, P2, P11		Eb1	300	515	V dc
				Eb1	400	690	V dc
		P7		Eb1	302	517	V dc
		P12					
5248	Deflection factor (1D2)	P1, P2, P11	Eb3= Eb2= 1,500 V dc	DF	102	138	V dc/in.
		P7	Eb3= Eb2= 2,000 V dc	DF	136	184	V dc/in.
		P12	Eb3= Eb2= 1,500 V dc	DF	96	144	V dc/in.
5248	Deflection factor (3D4)	P1, P2, P11	Eb3= Eb2= 1,500 V dc	DF	76	102	V dc/in.
		P7	Eb3= Eb2= 2,000 V dc	DF	100	138	V dc/in.
		P12	Eb3= Eb2= 1,500 V dc	DF	71	107	V dc/in.
5248	Deflection factor uniformity	All		- - -	- - -	- - -	- - -
5216	Cathode illumination	All		- - -	- - -	- - -	- - -
1331	Direct-interelectrode capacitance	All	g1 to all	Cg1	- - -	10.5	pF
			k to all	Ck	- - -	10.5	pF
			D1 to D2	C1D2	- - -	4.0	pF
			D3 to D4	C3D4	- - -	3.5	pF
			D1 to all	CD1	- - -	15.0	pF
			D2 to all	CD2	- - -	15.0	pF
			D3 to all	CD3	- - -	12.0	pF
			D4 to all	CD4	- - -	12.0	pF
1141	Pressure (implosion)	All		- - -	- - -	- - -	- - -
5111	Vibration	All		Width	- - -	1.0	mm
	Conformance inspection, part 1						
5201	Voltage breakdown	All		- - -	- - -	- - -	- - -
5201	Voltage breakdown (electrostatic types)	All		- - -	- - -	- - -	- - -
5206	Gas “cross”	P1, P2, P11	Ib3 = 100 µA dc (see note 7)	- - -	- - -	- - -	- - -
		P7	Ib3 = 200µA dc (see note 7)	- - -	- - -	- - -	- - -
		P12	Ib3 = 30µA dc (see note 7)	- - -	- - -	- - -	- - -
5106	Bulb, screen, and faceplate quality	All		- - -	- - -	- - -	- - -
- - -	Deflection plate cutoff	P1, P2, P7,P11	Ib3 = 10µA dc (max)(see note 3)	- - -	- - -	- - -	- - -
		P12	Ib3 = 5µA dc(max) (see note 3 and 4)	- - -	- - -	- - -	- - -

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Table 1. Testing and Inspection -Continued.

Method	Requirement or Test	Type	Conditions	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 1 - continued						
5223	Modulation	P7	Ib3 = 200 µA dc	ΔEc		55	V dc
		P12	Ib3 = 100 µA dc	ΔEc		40	V dc
5231	Spot position (electrostatic deflection)	P1, P2 P11,P12		- - -	- - -	15	mm
		P7		- - -	- - -	12	mm
5231	Spot displacement (leakage)	All		Displ	- - -	7	mm
5241	Grid-cutoff voltage	P1, P2 P11,P12		Ecl	-22.5	-67.5	V dc
		P7		Ecl	-30	-90	V dc
5251	Grid No. 1 leakage current	All		- - -	- - -	- - -	- - -
5251	Anode No. 2 leakage current	All		- - -	- - -	- - -	- - -
5221	Light output	P1	Eb2 = 1,500 V; Eb3 = 3,000 V; Ib3 = 100 µA dc	Light	25	- - -	ftL
		P11	Eb2 = 1,500 V; Eb3 = 3,000 V; Ib3 = 100 µA dc (see note 5)	Light	9	- - -	ftL
	Conformance inspection, part 2						
1301	Heater current	All		If	540	660	mA
5201	Electrode current (anode No. 1)	P1, P2, P11	Ib3 = 100 µA dc	Ib1	-50	10	µA dc
		P7	Ib3 = 200 µA dc	Ib1	-50	10	µA dc
		P12	Ib3 = 5 µA dc (see note 1)	Ib1	-50	10	µA dc
5101	Base alignment (electrostatic types)	All	+1D2, pin No. 5	- - -	- - -	- - -	- - -
5101	Side terminal alignment (electrostatic types)	All	+1D2	- - -	- - -	- - -	- - -
5201	Electrode current (anode No. 3)	P1, P2, P11	See note 2	Ib3	150	- - -	µA dc
5201	Electrode current (cathode)	P1, P2, P11	Ib3 = 100 µA dc	Ik	- - -	1,000	µA dc
		P7	Ib3 = 200 µA dc	Ik	- - -	1,500	µA dc
		P12	Ib3 = 5 µA dc (see note 1)	Ik	- - -	100	µA dc

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Table 1. Testing and Inspection -Continued.

Method	Requirement or Test	Type	Conditions	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 2 - Continued						
5101	Angle between traces	All		- - -	- - -	- - -	- - -
5101	Neck and base alignment (electrostatic types)	All		- - -	- - -	- - -	- - -
5216	Stray light emission (conventional types)	All	Eb2 = 2,200 V dc; Eb3 = 4,400 V dc	- - -	- - -	- - -	- - -
5221	Screens	P2		D1(1)	360	- - -	cB
		P7		D1(5)	400	- - -	cB
				G5 : 1	4	- - -	
		P12		cbf			
				D0.1 (E)	550	- - -	- - -
				D0.3 (E)	80	100	cB
5226	Line width “A” (electrostatic deflection)	P1, P2, P11	Ib3 = 100 µA dc (see note 6)	Width	- - -	0.75	mm
		P7	Ib3 = 200 µA dc (see note 6)	Width	- - -	0.80	mm
		P12	Ib3 = 5 µA dc (see notes 1 and 6)	Width	- - -	0.35	mm
5226	Line width “B” (electrostatic deflection)	P1, P2, P11	Ib3 = 100 µA dc (see note 8)	Width	- - -	0.90	mm
		P7	Ib3 = 200 µA dc (see note 8)	Width	- - -	1.00	mm
		P12	Ib3 = 5 µA dc (see notes 1 and 8)	Width	- - -	0.40	mm
5246	Focusing voltage at cutoff	P1, P2, P11		Eb1	345	515	V dc
		P7		Eb1	460	690	V dc
		P12		Eb1	345	517	V dc
5248	Deflection factor (1D2)	P1, P2, P11,P12		DF	127	173	V dc/in.
		P7		DF	170	230	V dc/in.
5248	Deflection factor (3D4)	P1, P2, P11,P12		DF	94	128	V dc/in.
		P7		DF	125	170	V dc/in
5251	Heater-cathode leakage current	All		- - -	- - -	- - -	- - -
1101	Secureness of base, cap, or insert	Al		- - -	- - -	- - -	- - -
1105	Permanence of marking			- - -	- - -	- - -	- - -
1111	Base pin solder depth			- - -	- - -	- - -	- - -

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Table 1. Testing and Inspection -Continued.

Method	Requirement or Test	Type	Conditions	Symbol	Limits Min	Limits Max	Unit
	Conformance inspection, part 3						
- - -	Life-test provisions	P1, P2, P7, P11	Group C; t = 500 hours (min) Eb3 = 4,400 V dc; Eb2 = 2,200 V dc; Ib3 = 30 µA dc	- - -	- - -	- - -	- - -
		P12	Group C; t = 500 hours (min) Eb3 = 4,400 V dc Eb2 = 2,200 V dc Ib3 = 5 µA dc	- - -	- - -	- - -	- - -
- - -	Life-test end points:						
5223	Modulation	P1, P2, P11,P12	Ib3 = 100 µA dc	ΔEc	- - -	40	V dc
		P7	Ib3 = 150 µA dc	ΔEc	- - -	55	V dc
5226	Line width “A” (electrostatic deflection)	P1, P2, P11	Ib3 = 100 µA dc	Width	- - -	0.75	mm
		P7	Ib3 = 150 µA dc	Width	- - -	0.86	mm
		P12	Ib3 = 5 µA dc	Width	- - -	0.35	mm
5226	Line width “B” (electrostatic deflection)	P1, P2, P11	Ib3 = 100 µA dc	Width	- - -	0.90	mm
		P7	Ib3 = 150 µA dc	Width	- - -	1.12	mm
		P12	Ib3 = 5 µA dc	Width	- - -	0.40	mm
1331	Direct-interelectrode capacitance	All	g1 to all	Cgl	- - -	10.5	pF
			k to all	Ck	- - -	10.5	pF
			D1 TO D2	C1D2	- - -	4.0	pF
			D3 TO D4	C3D4	- - -	3.5	pF
			D1 TO ALL	CD1	- - -	15.0	pF
			D2 TO ALL	CD2	- - -	15.0	pF
			D3 TO ALL	CD3	- - -	12.0	pF
			D4 TO ALL	CD4	- - -	12.0	pF
			(See Note 10)				

NOTES:

1 / The tube can be severely and permanently damaged if the current density on the screen is allowed to rise too high in static tests of this sort. For this reason, the length of time during which the screen is bombarded should be kept as short as possible compatible with the reading of specified current limits.

2 / Ib3 is to be measured with ΔEc = 40 V dc when cutoff is greater than -40 V dc and with Ecl = 0 when cutoff is less than -40 V dc.

3 / At specified conditions there shall be no pattern cutoff with the useful screen area.

4/ To prevent burning of the screen, the beam current should not be allowed to exceed that amount necessary to just produce a visible trace.

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NOTES - Continued:

5/ As measured by a 2 x 2 inch (50.80 x 50.80 mm) raster using a type 3 photronic cell without eye correction, calibrated in foot candles of illumination from a light source having color temperature of 2,700°K.

6/ The reflecting plates shall be returned to anode No. 2 through a 2.5 megohm resistance. Ib3, the beam current, shall be set as follows:

P1- - - - - - - - - Ib3 = 100 µA dc
P2 - - - - - - - - - Ib3 = 100 µA dc
P7- - - - - - - - - Ib3 = 200 µA dc
P11 - - - - - - - - - Ib3 = 100 µA dc
P12 - - - - - - - - - Ib3 = 5 µA dc

a/ The high frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus. The tube shall be observed for deflection defocusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of the trace (usually around the edges), bow-tying (irregular widths of any single line when observed at different points), bowing of trace other than that normally caused by curvature of bulb.

b/ This test for focus is to be made in addition to the line width measurements.

7/ This test to be performed at the conclusion of the holding period.

8/ The same conditions shall be set up as described in note 6 except that the connection of deflection elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 6 conditions.

An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 6.

9/ Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

10/ This test shall be performed during the initial production and once each succeeding 12- calendar months in which there is production. A sampling plan shall be used, with a sample of six tubes with an acceptance number of zero defects. The regular ‘12-calender month” sampling plan shall be reinstated after three consecutive samples have been accepted.

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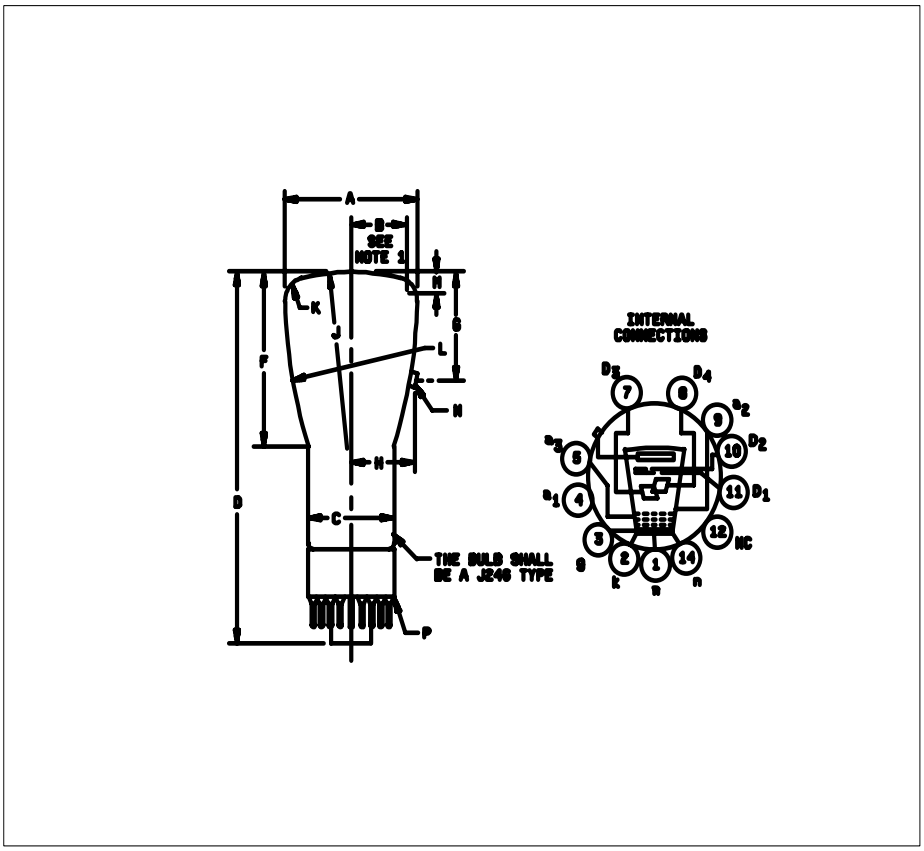

Dimensions in inches with metric equivalents (mm) in parentheses (see note b/)		
Ltr	Minimum	Maximum
Conformance inspection, part 2		
A	2.94 (74.7)	3.06 (77.7)
B	1.37 (34.8)	
C	1.94 (49.3)	2.06 (52.3)
D	9.75 (247.6)	10.25 (260.4)
F	3.75 (95.2)	4.00 (101.6)
G	1.50 (38.1)	2.00 (50.8)

Table 1 continued.	
Conformance inspection, part 3 (See note 10/)	
N	Bulb contact: J1-22
P	Base: B12-37
Reference dimensions (see note c/)	
H	1.62 (41.1)
J	8.00 (203.2) R
K	.25 (6.4) R
L	12.44 (316.0) R
M	.35 (8.9)

NOTES:
a/ The minimum useful screen radius shall not be less than 1.375 inches (34.92 mm).
b/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
c/ Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3JP1, 3JP2, 3JP7, 3JP11, AND 3JP12.

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